XCELERATE | Python Basic

Duration - 2 Days

Header	Description
Course Objective	By the end of the program participants will be able to: • Understand the core fundamentals of Python, including identifiers, keywords, data types, operators, and program flow control. • Work confidently with Python's built-in data structures such as strings, lists, tuples, sets, and dictionaries, and apply key operations on them. • Apply conditional and iterative logic using Python's flow control statements to build structured programs. • Use Python functions effectively, including user-defined functions, lambda expressions, recursion, decorators, and built-in functional constructs like map, filter, and reduce. • Work with modules and packages, import mechanisms, the math and random libraries, and understand Python's modular programming structure. • Handle user input and output, perform type conversions, format strings, and write interactive console programs. • Understand and apply regular expressions for validation, searching, pattern matching, and text processing using Python's re module. • Develop logic-building skills through extensive pattern-based programs (stars, numbers, alphabets, pyramids, diamonds).
Lab Setup	• Each participant to carry laptop with internet connection

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Detailed Syllabus

1. Introduction to Python

- What is Python
- History of Python
- Why Python? (Beginner-friendly, simple syntax)
- Applications of Python (Web, ML, AI, Data Analysis, IoT, etc.)

2. Language Fundamentals

- Identifiers
- Keywords / Reserved Words
- Constants
- Escape Characters
- Comments
- Input / Output Basics

3. Data Types in Python

- Numbers (int, float, complex)
- Boolean
- String & String Operations
- Bytes & Bytearray
- List
- Tuple
- Set & Frozenset
- Dictionary
- None Type
- Mutability vs Immutability
- Type Casting (int(), float(), str(), complex(), bool())

4. Operators in Python

- Arithmetic Operators
- Relational / Comparison Operators
- Equality Operators
- Logical Operators
- Bitwise Operators
- Assignment & Compound Assignment
- Ternary Operator
- Identity Operators (is, is not)
- Membership Operators (in, not in)
- Operator Precedence

5. Flow Control Statements

- Conditional Statements
  - if
  - if–else
  - if–elif–else
- Iterative Statements
  - for loop
  - while loop
  - Nested loops
  - Loop else block
- Transfer Statements
  - break
  - continue
  - pass
  - del

6. String Handling & Operations

- Indexing & Slicing
- String Methods (find, replace, count, split, join, strip, case conversions)
- Character-Type Tests (isalnum, isalpha, isdigit, etc.)
- Formatting Strings (format(), {})
- Common String Programs (reverse, character count, merge, patterns)

7. List, Tuple, Set & Dictionary (Collections)

- List: creation, indexing, slicing, methods (append, extend, insert, remove, pop, sort, reverse)
- Tuple: immutability, tuple operations, packing/unpacking
- Set: uniqueness, set operations, union/intersection/difference
- Frozenset: immutable set type
- Dictionary: key-value pairs, creation, updating, deletion, dictionary methods

8. Functions in Python

- Defining functions
- return statement
- Types of arguments
- Variable scope (local, global)
- lambda functions
- filter(), map(), reduce()
- Function aliasing
- Nested functions
- Decorators

9. Modules & Packages

- Creating and importing modules
- import, from … import
- Aliasing modules & members
- dir() function
- Special variable __name__

- Using math module
- Using random module
- Package directory structure

10.Generators & Iterators

- Generator functions
- yield
- Advantages for memory optimization
- Generator expressions

11.Pattern Programming

- Star Patterns
- Number Patterns
- Alphabet Patterns
- Pyramid / Diamond / Mirror Patterns (Patterns 1–100)

12.Regular Expressions (re Module)

- Character classes
- Predefined character sets (\d, \w, \s, etc.)
- Quantifiers
- Anchors (^, $)
- re functions: match, search, finditer, findall, fullmatch, split, sub, subn
- Validation & extraction programs (email, phone numbers, identifiers)

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